Pythagoras in 3D

Name _____ **Date** _____ Calculator allowed Version 1

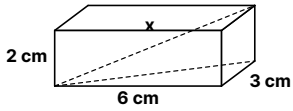
KEY WORDS diagonal · slant height · right-angled triangle · hypotenuse · cuboid · cone · pyramid

A The diagonal of a cuboid ●○○ ANSWER ON THE SHEET

Remember: Find a right-angled triangle inside the solid. For the long diagonal of a box, use the diagonal of the base first: $x = \sqrt{l^2 + w^2 + h^2}$.

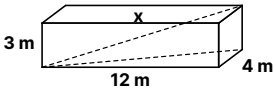
1. Find the length of the diagonal marked x. Give answers to 1 decimal place where needed.

a)



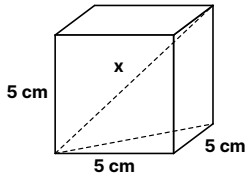
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b)



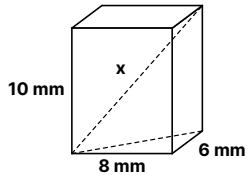
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c)



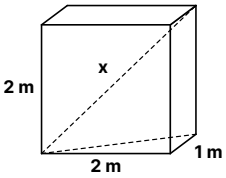
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d)



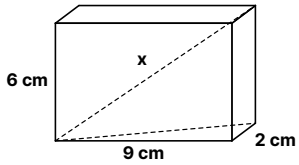
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e)



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f)



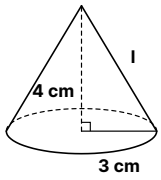
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B Cones ●●○ ANSWER ON THE SHEET

Remember: The radius, the height and the slant height of a cone make a right-angled triangle. The slant height is the hypotenuse.

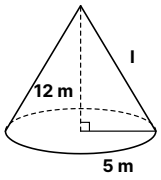
2. Find the length marked with a letter.

a)



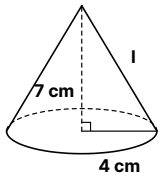
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b)



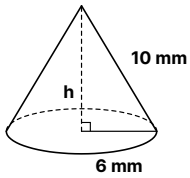
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c)



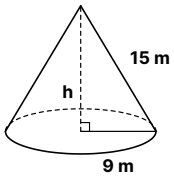
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d)



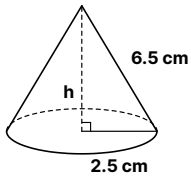
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e)



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f)



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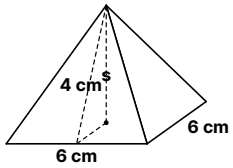
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C Pyramids ●●● WORK IN YOUR BOOK

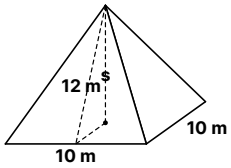
Remember: The slant height of a face uses half the base edge. A slanting edge uses half the base diagonal. Draw the triangle you are using.

3. Each pyramid has a square base. Find the length marked with a letter.

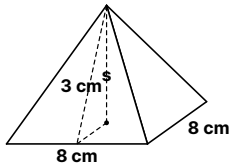
a)



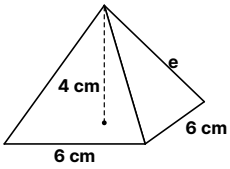
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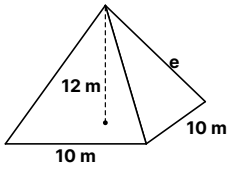
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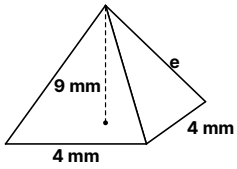
d)



e)



f)



D Problems ●●● WORK IN YOUR BOOK

4. A pencil case is a box 12 cm by 8 cm by 5 cm. Find the length of the longest pencil that fits inside, to 1 decimal place. Will a 15 cm pencil fit?

5. A cylindrical tube has a diameter of 6 cm and a height of 20 cm. Find the length of the longest straight rod that fits inside it, to 1 decimal place.

6. A tent pole is 2.4 m tall. A guy rope runs from the top of the pole to a peg 1.5 m from its base. How long is the rope, to 2 decimal places?

7. A wheelchair ramp rises 0.6 m over a horizontal distance of 2.4 m. Find the length of the sloping surface, to 2 decimal places.

★ Extension FOR EARLY FINISHERS

8. Show that the long diagonal of a cube with edges of 4 cm is $4\sqrt{3}$ cm, and explain why the diagonal of any cube is $\sqrt{3}$ times its edge.

9. A box is 3 m by 4 m by 12 m. How much longer is its long diagonal than its longest edge?

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Pythagoras in 3D

Name _____ **Date** _____ Calculator allowed **Version 2**

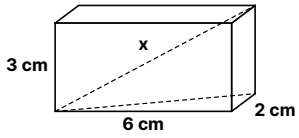
KEY WORDS diagonal · slant height · right-angled triangle · hypotenuse · cuboid · cone · pyramid

A The diagonal of a cuboid ●○○ ANSWER ON THE SHEET

Remember: Find a right-angled triangle inside the solid. For the long diagonal of a box, use the diagonal of the base first: $x = \sqrt{l^2 + w^2 + h^2}$.

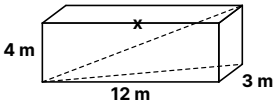
1. Find the length of the diagonal marked x. Give answers to 1 decimal place where needed.

a)



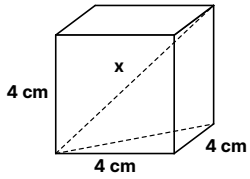
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b)



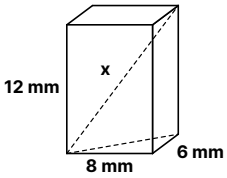
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c)



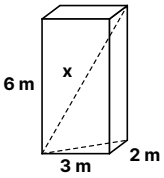
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d)



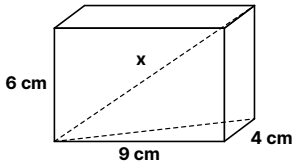
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e)



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f)



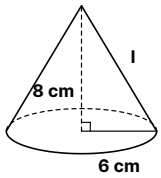
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B Cones ●●○ ANSWER ON THE SHEET

Remember: The radius, the height and the slant height of a cone make a right-angled triangle. The slant height is the hypotenuse.

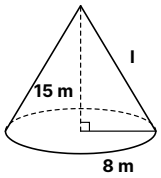
2. Find the length marked with a letter.

a)



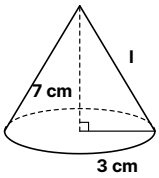
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b)



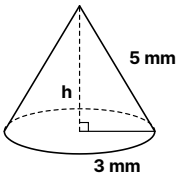
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c)



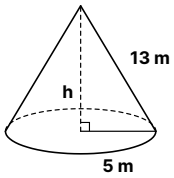
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d)



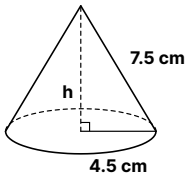
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e)



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f)



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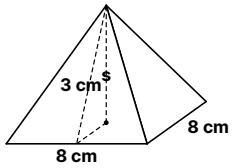
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C Pyramids ●●● WORK IN YOUR BOOK

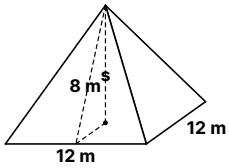
> **Remember:** The slant height of a face uses half the base edge. A slanting edge uses half the base diagonal. Draw the triangle you are using.

3. Each pyramid has a square base. Find the length marked with a letter.

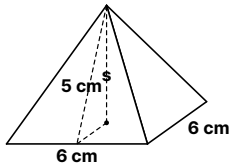
a)



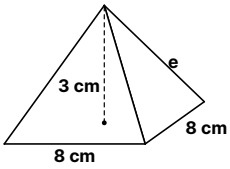
b)



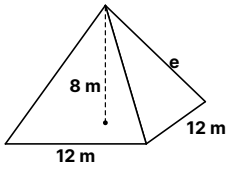
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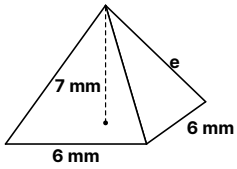
d)



e)



f)



D Problems ●●● WORK IN YOUR BOOK

4. A pencil case is a box 15 cm by 10 cm by 6 cm. Find the length of the longest pencil that fits inside, to 1 decimal place. Will a 19 cm pencil fit?

5. A cylindrical tube has a diameter of 8 cm and a height of 24 cm. Find the length of the longest straight rod that fits inside it, to 1 decimal place.

6. A tent pole is 2.8 m tall. A guy rope runs from the top of the pole to a peg 2 m from its base. How long is the rope, to 2 decimal places?

7. A wheelchair ramp rises 0.8 m over a horizontal distance of 3 m. Find the length of the sloping surface, to 2 decimal places.

★ Extension FOR EARLY FINISHERS

8. Show that the long diagonal of a cube with edges of 6 cm is $6\sqrt{3}$ cm, and explain why the diagonal of any cube is $\sqrt{3}$ times its edge.

9. A box is 2 m by 6 m by 9 m. How much longer is its long diagonal than its longest edge?

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